.Gatekeeper

Decision support for the European Health Data Space

AIM Healthcare Hackathon Vienna 2026, Track 3. Oguz Nacar and Arda Sarikaya.

Reliable health data is expensive and complicated to access

EUR 4,000

Standard processing fee for one application in Germany.

Paid before anyone knows whether the request can be granted.

Medical research depends on reliable health data.

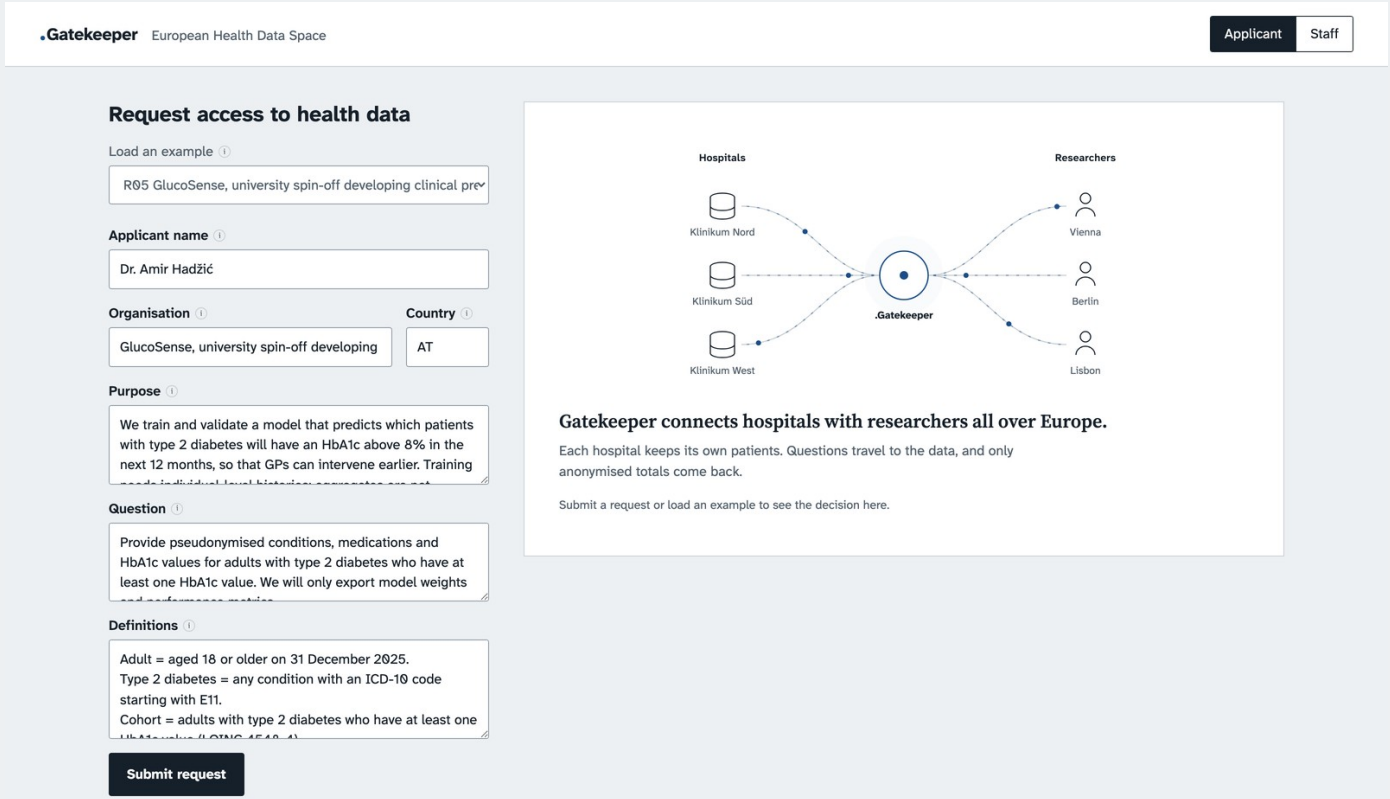
Every application goes through a Health Data Access Body under Regulation (EU) 2025/327.

Applicants write the request, wait, and find out late which article blocks it.

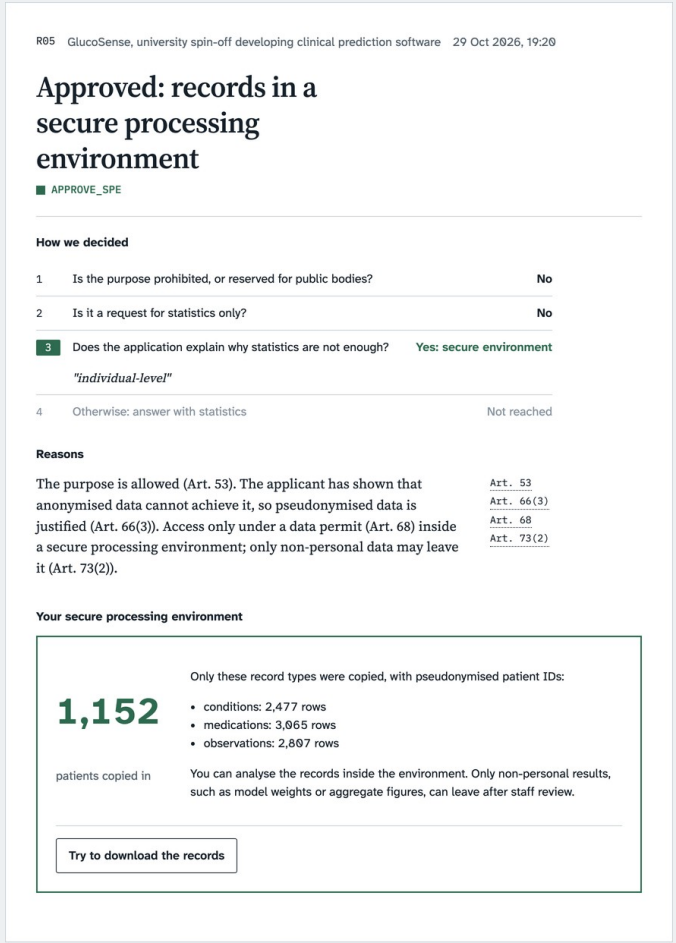
That is why we created Gatekeeper.

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Applicant interface: submit, then see the decision



Request form: purpose, question, definitions.



Decision notice: APPROVE_SPE with Art. 53, 66(3), 68, 73(2).

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AI decision engine: structured, consistent, explainable

1

Each request is analyzed

Purpose, question and definitions are read as structured fields.

2

Relevant requirements are checked

Chapter IV of Regulation (EU) 2025/327, rule by rule.

3

An explainable recommendation

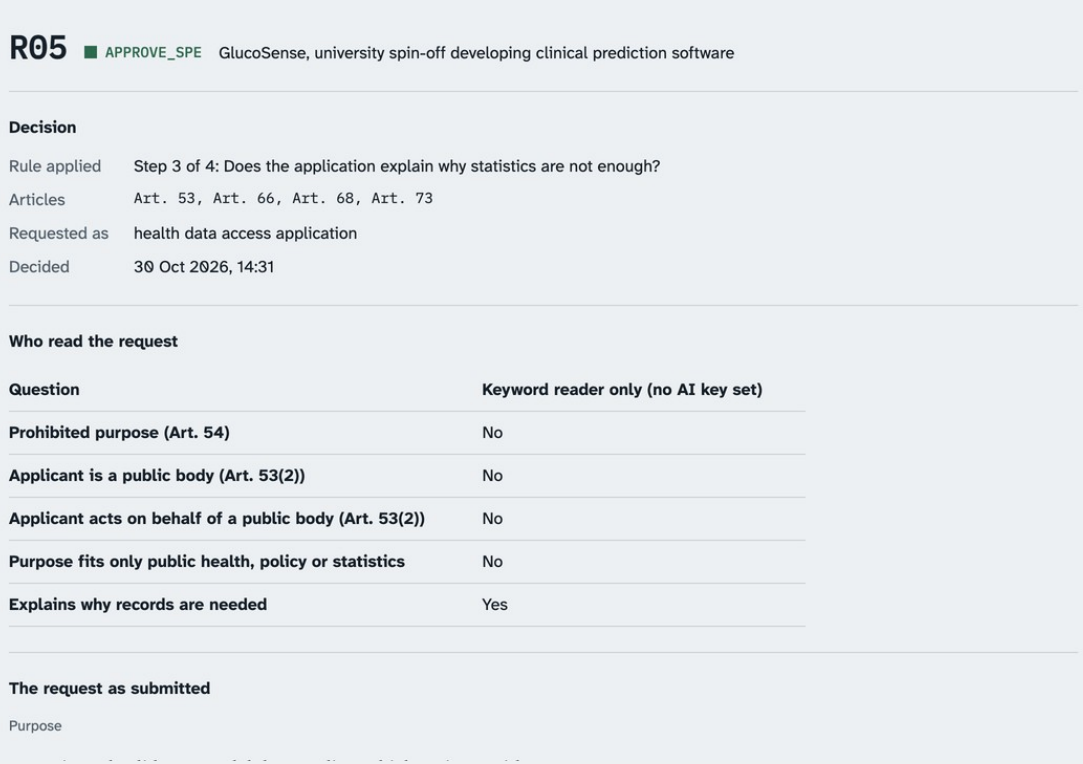
Every outcome names the rule applied and cites the articles.

Three possible outcomes

`REJECT` `APPROVE_STATISTICAL` `APPROVE_SPE`

Specialized decision models instead of a free-form LLM answer: the same request gets the same decision.

Staff interface and federated data



R05 ■ APPROVE_SPE GlucoSense, university spin-off developing clinical prediction software

Decision

Rule applied	Step 3 of 4: Does the application explain why statistics are not enough?
Articles	Art. 53, Art. 66, Art. 68, Art. 73
Requested as	health data access application
Decided	30 Oct 2026, 14:31

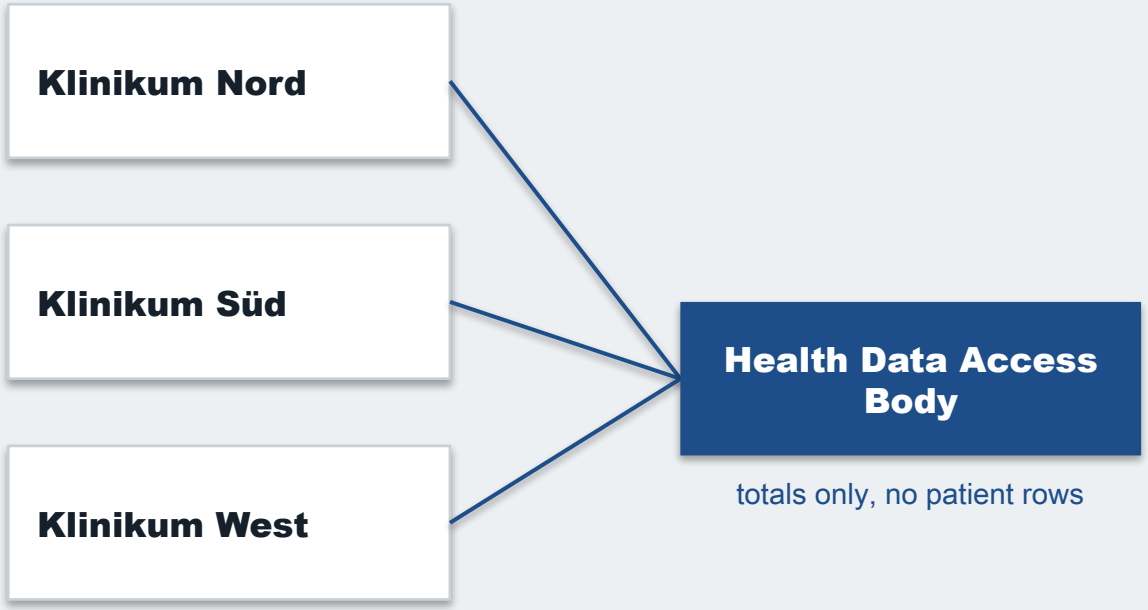
Who read the request

Question	Keyword reader only (no AI key set)
Prohibited purpose (Art. 54)	No
Applicant is a public body (Art. 53(2))	No
Applicant acts on behalf of a public body (Art. 53(2))	No
Purpose fits only public health, policy or statistics	No
Explains why records are needed	Yes

The request as submitted

Purpose

Staff case file: rule applied, articles, who read the request, numbers.



Queries run inside each hospital. Patient rows never leave; only totals are merged centrally.

Opted-out patients excluded. Counts under 10 shown as <10.

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Results on the 10 sample requests

10/10	6/6	0	10/10	12.7 s
correct decisions	correct numbers	privacy leaks	consistent over 3 runs	slowest request

Measured on the 10 sample requests, synthetic data.

Decision accuracy, statistical correctness, privacy protection, consistency, processing time.

Vision and limitations

Today

Prototype runs the full workflow on synthetic hospital data.

Three hospital databases in one process; in production each sits behind its own service.

Cases live in memory; no login; runs locally.

Future work

Integration with real European healthcare infrastructure.

Help research institutions access and analyze health data more reliably, affordably, and securely.

Making European health research safer, smarter, and more accessible.